3.1, 10, 14, 24, 25

Problem 3.1. Use Table 3.1 to compute the temperatures of solid A and solid B when $q_A = 1$. Then compute both temperatures when $q_A = 60$. Express your answers in terms of ϵ/k, and then in kelvins assuming that $\epsilon = 0.1$ eV.

$$\frac{1}{T} = \left(\frac{\partial S}{\partial U}\right)_{N,V} \qquad q_A = 1 \qquad \frac{S_A}{k} = 5.7 \qquad \frac{S_B}{k} = 186.4$$

$$U = q \cdot \epsilon \Rightarrow U(q=1) = 1 \cdot \epsilon$$

$$T_A = \frac{\partial U_A}{\partial S_A} = \frac{\Delta U_A}{\Delta S_A} = \frac{1\epsilon - 0\epsilon}{5.7 k_B - 0} = \frac{\epsilon}{5.7}$$

$$T_B = \frac{\Delta U_B}{\Delta S_B} = \frac{99\epsilon - 100\epsilon}{(186.4 - 187.5)k_B} = \frac{\epsilon}{1.1 k_B}$$

$$T_A < T_B$$

$$T_A = \frac{\epsilon}{5.7 \cdot k_B}$$

$$\epsilon = 0.1\,eV$$

$$T_A = \frac{0.1\,eV}{5.7 \cdot 8.6 \cdot 10^{-5}\,eV/K}$$

$$\rightarrow \boxed{T_A = 204\,K}$$

$$T_B = \frac{0.1}{1.1 \cdot 8.6 \cdot 10^{-5}}$$

$$\rightarrow \boxed{T_B = 1057\,K}$$

$$k_B = 1.38 \cdot 10^{-23}\,J/K$$

$$1eV = 1.6 \cdot 10^{-19}\,J$$

$$k_B = 8.6 \cdot 10^{-5}\,eV/K$$

$$T_A = \frac{\epsilon}{(159.1 - 157.4)k_B} = \boxed{\frac{\epsilon}{1.7\,k_B}}$$

$$T_B = \frac{-\epsilon}{(105.3 - 107.0)k_B} = \boxed{\frac{\epsilon}{1.7 k_B}}$$

$T_A = T_B$ ← at equilibrium

$$\boxed{T_A = 684\,K}$$

Problem 3.10. An ice cube (mass 30 g) at 0°C is left sitting on the kitchen table, where it gradually melts. The temperature in the kitchen is 25°C.

(a) Calculate the change in the entropy of the ice cube as it melts into water at 0°C. (Don't worry about the fact that the volume changes somewhat.)

(b) Calculate the change in the entropy of the water (from the melted ice) as its temperature rises from 0°C to 25°C.

(c) Calculate the change in the entropy of the kitchen as it gives up heat to the melting ice/water.

(d) Calculate the net change in the entropy of the universe during this process. Is the net change positive, negative, or zero? Is this what you would expect?

a. latent heat — ice → water $Q = 333.7\ \frac{J}{g} \cdot 30\,g = L_H \cdot m$

$$\Delta S = \frac{Q}{T} = \frac{333.7 \cdot 30}{273.15} = 36.6\ \frac{J}{K}$$

$$dQ = T\,dS$$

$$\frac{dQ}{T} = dS \quad \text{(infinitesimal change)}$$

$$\frac{Q}{T} = \Delta S \quad \text{(to a macroscopic change)}$$

b. $$dS = \frac{Q}{T} = \frac{C_v\,dT}{T}$$

heat capacity — specific heat capacity

$C_v = m\,c_v$

mass — $4.186\ \frac{J}{gK}$

$$\Delta S = \int dS = \int \frac{C_v}{T} dT = mC_v \int_{273}^{298} \frac{dT}{T} = mC_v \ln T \Big|_{T_i}^{T_f} = mc\left(\ln T_f - \ln T_i\right) = mc \ln\left(\frac{T_f}{T_i}\right)$$

$$\Delta S = 30\,g \cdot 4.186 \frac{J}{g\,K} \cdot \ln\left(\frac{298}{273}\right)$$

$$= 11\ J/K$$

3.25

$$\Omega(N,q) \approx \left(\frac{q+N}{q}\right)^q \left(\frac{q+N}{N}\right)^N$$

a) Find S

$$\frac{S(N,q)}{k_B} = \ln\Omega = q\ln\left(\frac{q+N}{q}\right) + N\ln\left(\frac{q+N}{N}\right)$$

$$\frac{S(N,q)}{k_B} = q\ln\left(1+\frac{N}{q}\right) + N\ln\left(\frac{q}{N}+1\right)$$

b) Find T.

$U = q\cdot\epsilon \Rightarrow q = \frac{U}{\epsilon}$

$$\frac{1}{T} = \frac{\partial S}{\partial U} \longleftarrow \frac{S(N,U)}{k_B} = \frac{U}{\epsilon}\ln\left(1+\frac{N\cdot\epsilon}{U}\right) + N\ln\left(\frac{U}{\epsilon\cdot N}+1\right)$$

$$\frac{1}{T\cdot k_B} = \frac{1}{\epsilon}\ln\left(1+\frac{N\cdot\epsilon}{U}\right) + \frac{U}{\epsilon}\cdot\frac{U}{U}\cdot\frac{1}{1+\frac{N\epsilon}{U}}\cdot(-1)\frac{N\epsilon}{U^2} + N\cdot\frac{N\epsilon}{NU}\frac{1}{\frac{U}{\epsilon N}+1}\cdot\frac{1}{\epsilon N}$$

$$\underbrace{-\frac{U^2}{\epsilon}\frac{1}{U+N\epsilon}\cdot\frac{N\epsilon}{U^2} + N^2\epsilon\cdot\frac{1}{U+N\epsilon}\cdot\frac{1}{\epsilon N}}_{=0}$$

$$\frac{1}{k_B T} = \frac{1}{\epsilon} \cdot \ln\left(1 + \frac{N\epsilon}{u}\right)$$

$$T = \frac{\epsilon}{k_B} \cdot \frac{1}{\ln\left(1 + \frac{N\epsilon}{u}\right)}$$

c) Find u, then C

$$\frac{\epsilon}{k_B T} = \ln\left(1 + \frac{N\epsilon}{u}\right)$$

$$e^{\frac{\epsilon}{k_B T}} = 1 + \frac{N\epsilon}{u}$$

$$e^{\frac{\epsilon}{k_B T}} - 1 = \frac{N\epsilon}{u}$$

$$\boxed{u = \frac{N\epsilon}{e^{\frac{\epsilon}{k_B T}} - 1}}$$

$$\Rightarrow C = \frac{du}{dT} = \frac{N\epsilon^2}{k_B T^2} \cdot \frac{e^{\epsilon/k_B T}}{\left(e^{\epsilon/k_B T} - 1\right)^2}$$

d) Show C at high temps $T \to \infty$

$$C = \frac{dU}{dT} = \frac{N\epsilon^2}{k_B T^2} \cdot \frac{e^{\epsilon/k_B T}}{\left(e^{\epsilon/k_B T} - 1\right)^2}$$

$e^x \approx 1 + x$ for small x

$$C = \frac{N\epsilon^2}{k_B T^2} \cdot \frac{1 + \frac{\epsilon}{k_B T}}{\left(1 + \frac{\epsilon}{k_B T} - 1\right)^2} \qquad \frac{\epsilon}{k_B T} \to 0$$

$$C = \frac{N\epsilon^2}{k_B T^2} \cdot \frac{k_B^2 T^2}{\epsilon^2}$$

$$C = N k_B \checkmark$$

What do we expect?

from Einstein solid at high temp

$$U = N k_B T$$

$$C = \frac{dU}{dT} = N k_B \checkmark$$

$e^x \approx 1 + x$

$e^x \approx 1 + x + \frac{x^2}{2} + \frac{x^3}{6}$

R's off the dome guess

$\Rightarrow \frac{1}{t} = \frac{\epsilon}{k_B T}$

e) $$C = \frac{N\epsilon^2}{k_B T^2} \cdot \frac{e^{\epsilon/k_B T}}{\left(e^{\epsilon/k_B T} - 1\right)^2}$$

$$t = \frac{k_B T}{\epsilon} \Rightarrow T = \frac{\epsilon t}{k_B}$$

$$C = \frac{N\epsilon^2}{k_B} \cdot \frac{k_B^2}{\epsilon^2 t^2} \cdot \frac{e^{1/t}}{\left(e^{1/t} - 1\right)^2}$$

$$C = N k_B \cdot \frac{e^{1/t}}{t^2 \left(e^{1/t} - 1\right)^2}$$

$$\frac{C}{N k_B} = \underbrace{\frac{1}{t^2} \cdot \frac{e^{1/t}}{\left(e^{1/t} - 1\right)^2}}_{\text{plot}}$$

$y(t)$

